

MetroFlow AI

Urban Traffic Congestion Analyzer & Offline Signal Timing Recommendation Platform

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1. Executive Summary

MetroFlow AI is an offline urban traffic analytics and signal timing optimization platform designed specifically for municipal traffic engineers, smart city consultants, and urban transportation authorities. Urban intersections frequently suffer from severe delay and congestion caused by outdated, static signal timing plans that fail to reflect modern traffic fluctuations. MetroFlow AI ingests historical traffic volume data, road metadata, and incident logs to automatically detect bottleneck patterns, compute optimal offline green-split and offset adjustments using queuing theory, and generate simulated "before vs. after" performance metrics alongside plain-language engineering rationales. By replacing slow, manual signal retiming workflows with an intelligent advisory system, MetroFlow AI reduces corridor travel times, decreases vehicle emissions, and optimizes municipal infrastructure without requiring real-time signal control hardware modifications.

2. Product Vision

To become the global standard for data-driven municipal traffic management, empowering every city to eliminate preventable gridlock through intelligent, explainable, and accessible signal timing optimization.

3. Mission Statement

Deliver an offline traffic analytics platform that transforms historical vehicle count data into actionable, safe, and simulated signal timing recommendations, enabling traffic engineers to optimize corridor efficiency within minutes rather than months.

4. Problem Statement

Urban traffic congestion costs cities billions annually in lost productivity, excess fuel consumption, and heightened carbon emissions. Municipal traffic signal timing plans are traditionally updated only once every 3 to 5 years due to the high cost and manual labor required for traffic counting, micro-simulation setup (e.g., VISSIM, Synchro), and manual plan generation. Consequently, signal controllers run static timing programs that do not match current traffic flows, causing artificial gridlock, excessive vehicle idling, and uncoordinated corridor progression. Municipalities require an offline decision-support tool that ingests available volume data, identifies peak bottleneck patterns, and recommends optimized green split schedules accompanied by explainable metrics and safety guardrails.

5. Goals

- Automate Bottleneck Detection:** Reduce the time required to identify peak traffic congestion windows and corridor bottlenecks from weeks to under 30 seconds.
- Optimize Green Split & Coordination:** Generate offline signal timing plans (cycle lengths, green splits, and offsets) that yield an estimated 12%–25% reduction in average intersection delay based on historical flow data.
- Explainable Recommendations:** Provide plain-language narrative explanations powered by AI for every recommended signal change to build engineer trust.

- **Low-Barrier Simulation:** Supply lightweight "before vs. after" queue and delay simulation estimates without requiring complex, proprietary micro-simulation software setups.
- **Ensure Zero Operational Risk:** Restrict functionality strictly to offline recommendations and exportable plans, avoiding direct signal controller override or safety hazard creation.

6. Non-Goals

- **No Real-Time Signal Control:** The system will NOT directly actuate or interface with live traffic signal controllers in real time.
- **No Autonomous Enforcement:** The platform will NOT perform automated speed/red-light enforcement, vehicle identification, or driver profiling.
- **No Full Physics Micro-Simulation Engine Replacement:** The system will NOT replace heavy microscopic traffic simulation platforms like VISSIM or AIMSUN for complex civil engineering structural redesigns; it uses lightweight queuing models for timing validation.
- **No Hardware Manufacturing:** The solution is entirely software-based and will not produce custom sensor hardware.

7. Target Users

- **Primary:** Municipal Traffic Engineers, Traffic Operations Center (TOC) Managers.
- **Secondary:** Smart City Consultants, Transportation Planning Agencies, Regional Councils of Governments (COGs).
- **Tertiary:** Civil Engineering Contractors, Municipal Operations Directors.

8. User Personas

Persona 1: Senior Traffic Engineer (Primary) Name: Marcus Vance, PE Role: Senior Traffic Operations Engineer, City Department of Transportation Demographics: Age 44, M.S. in Civil Engineering, 18 years of experience. Goals: Clear corridor bottlenecks, lower intersection level-of-service (LOS) delay grades, defend signal retiming decisions to city council. Pain Points: Synchro software takes weeks to model and calibrate; current static timing plans are 4 years old; lacks time to manually parse raw loop-detector CSV logs.	Persona 2: Smart City Consultant (Secondary) Name: Elena Rostova Role: Mobility Practice Lead, Urban Infrastructure Advisory Demographics: Age 35, M.S. in Urban Planning & Data Science, 10 years of experience. Goals: Deliver data-backed signal optimization proposals to municipal clients quickly; demonstrate quantifiable ROI and emissions reduction. Pain Points: Municipal traffic data is fragmented; clients lack budget for expensive real-time adaptive signal hardware (e.g., SCATS/SCOOT); needs clear visual reporting for non-technical city leadership.
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9. Stakeholder Analysis

Stakeholder Group	Interest Level	Power / Influence	Core Needs & Expectations
Municipal Traffic Engineers	High	High	Accuracy, adherence to MUTCD standards, exportable timing sheets, clear simulation assumptions.
City DOT Leadership	High	High	Congestion reduction, cost efficiency, public safety compliance, clear reporting metrics.
Commuter Public & Transit	High	Low	Reduced travel time, fewer stops, predictable signal progression, lower fuel consumption.
Smart City Consultants	High	Medium	Rapid analysis turnaround, white-label export capabilities, multi-intersection scenario modeling.
IT & GIS Departments	Medium	Medium	Easy CSV/JSON data ingestion, secure web access, zero live infrastructure integration risks.

10. Pain Points

- **Static Timing Inflexibility:** Intersections run static 24-hour plans that fail during unannounced volume surges or seasonal shifts.
- **Data Overwhelm vs. Action Gap:** Cities collect millions of vehicle count data points from loop detectors and cameras, but lack tools to automatically synthesize this into timing parameters.
- **High Retiming Costs:** Traditional retiming projects cost \$3,000–\$5,000 per intersection and take 6–12 months to execute manually.
- **Black-Box AI Skepticism:** Traffic engineers reject AI systems that provide raw numbers without engineering rationale or explicit mathematical assumptions.
- **Complex Modeling Overhead:** Existing simulation software requires specialized expertise, long render times, and manual input parameter mapping.

11. Existing Workflow

1. **Manual Data Collection:** Portable pneumatic tubes or historical inductive loop logs record vehicle counts over a 7-day period.
2. **Data Sanitization:** Engineers manually clean corrupted CSVs in Excel.
3. **Model Construction:** Data is manually imported and mapped into traditional traffic software (e.g., Synchro/HCS).
4. **Manual Calibration:** Engineer manually adjusts phase splits, yellow intervals, and all-red clearance times over several weeks.
5. **Field Deployment:** Approved parameters are manually keyed into physical signal controller cabinets on site.
6. **Post-Evaluation:** Minimal or no post-implementation validation occurs due to budget constraints.

12. Proposed Workflow

1. **Data Ingestion:** User uploads historical intersection volume logs (CSV/JSON), road metadata (lanes, speed limits, turning ratios), and optional incident reports into MetroFlow AI.
2. **Automated Analytics:** Platform automatically parses counts, identifies peak hourly patterns, computes volume-to-capacity (V/C) ratios, and highlights specific corridor bottlenecks.

3. **Optimization Engine Execution:** Web-based queuing model calculates optimized cycle length, green splits, and offset values tailored to peak demand periods.
4. **Simulated Validation:** System runs an instant "before vs. after" simulation displaying predicted reductions in queue length, average delay (seconds/vehicle), and Level of Service (LOS) grade improvements.
5. **AI Rationale & Plan Export:** System generates a structured timing recommendation sheet accompanied by an AI-generated engineering narrative explaining the changes and explicitly stating underlying assumptions.
6. **Controller Cabinet Update:** Traffic engineer reviews, approves, and downloads standard timing controller parameters for scheduled field programming.

13. User Stories (Minimum 30)

User Story Index: The following 30 comprehensive user stories cover end-to-end functionality across Data Ingestion, Congestion Analytics, Signal Optimization, Simulation, and AI Rationale Reporting.

ID	User Role	Requirement ("I want to...")	Benefit ("So that...")
US-01	Traffic Engineer	upload intersection vehicle count data in standard CSV format	I do not have to reformat raw loop detector outputs manually.
US-02	Smart City Consultant	define road metadata (number of lanes, speed limits, storage bay lengths)	the analyzer accurately calculates intersection capacity.
US-03	Analyst	input turning movement counts (Left, Through, Right) for each approach	phase green time distribution reflects real movement demands.
US-04	Traffic Engineer	attach historical incident logs (accidents, roadwork)	non-recurrent congestion anomalies are filtered out from baseline timing.
US-05	System Admin	automatically validate uploaded dataset formats	corrupt or missing timestamp records are flagged prior to analysis.
US-06	Analyst	save multi-intersection corridor datasets into project folders	I can analyze arterial networks as a unified system.
US-07	Traffic Engineer	view a heat map of intersection delay across different times of day	I can rapidly identify peak congestion windows.
US-08	TOC Manager	automatically rank top corridor bottlenecks by V/C ratio	I can prioritize retiming intervention budgets.
US-09	Analyst	inspect directional traffic flow imbalances during rush hours	asymmetric split plans can be generated.
US-10	Smart City Consultant	compare traffic volumes across days of the week	I can determine whether separate weekend vs. weekday timing plans are needed.
US-11	Traffic Engineer	view queue spillback risk scores on short turn lanes	blockage of through-lanes can be prevented.
US-12	Executive	view aggregate city-wide congestion trend metrics	I can report smart city progress to municipal leadership.
US-13	Traffic Engineer	calculate optimal green split distributions based on approach saturation	green time allocation matches real demand.
US-14	Senior Engineer	enforce minimum pedestrian crossing time rules	pedestrian safety is never compromised.
US-15	Traffic Engineer	suggest signal coordination offsets across sequential corridor intersections	green-wave platooning is maximized.
US-16	Analyst	recommend optimal cycle lengths (e.g., 90s, 120s, 150s) for peak bands	cycle overhead is minimized without causing overflow queues.
US-17	Traffic Engineer	manually override and adjust suggested green splits	I can incorporate local field knowledge into the plan.
US-18	Consultant	lock specific phase parameters while re-optimizing others	transit or railroad preemption phases remain unchanged.
US-19	Traffic Engineer	view a "before vs. after" comparative table of average delay per vehicle	I can evaluate the performance gain of the proposed plan.
US-20	Analyst	view simulated queue length reductions in meters/feet per lane	I can verify queues will not spill into upstream intersections.
US-21	TOC Manager		statutory reporting criteria are met.

ID	User Role	Requirement ("I want to...")	Benefit ("So that...")
		see expected Level of Service (LOS A-F) transitions for each approach	
US-22	Environmental Analyst	estimate fuel savings and CO2 emission reductions	sustainability metrics can be tracked.
US-23	Senior Engineer	view all underlying mathematical assumptions used in queuing models	I can validate simulation integrity.
US-24	Analyst	run sensitivity stress tests (+10% traffic surge) on proposed plans	I can verify plan robustness under unpredictable demand.
US-25	Traffic Engineer	receive an AI-generated plain-language summary of timing adjustments	I can quickly understand optimization logic.
US-26	Consultant	export official PDF signal retiming reports	I can submit professional deliverables directly to municipal clients.
US-27	Operations Lead	export controller-ready timing sheets (in CSV/ NEMA format)	technicians can easily program physical hardware on site.
US-28	Senior Engineer	cite relevant traffic engineering standards (HCM/MUTCD) in summaries	compliance is transparent and defensible.
US-29	Non-technical Manager	view executive summary narratives stripped of complex jargon	I can brief non-technical city council members easily.
US-30	System Auditor	verify explicit offline disclaimers on every export	legal safety liability guardrails are strictly enforced.

14. User Journey

- 1. Landing & Authentication:** Marcus logs into MetroFlow AI via secure municipal SSO.
- 2. Project Setup & Data Upload:** Marcus creates project "Main Street Corridor Retiming" and uploads a 7-day intersection traffic volume CSV alongside a road geometry metadata file.
- 3. Data Verification & Parsing:** System parses 100,000+ vehicle counts, verifies data integrity, and generates a visual baseline corridor overview showing peak congestion occurring weekdays between 07:45 AM - 09:15 AM.
- 4. Bottleneck Identification:** System flags Intersection #4 (Main St & 5th Ave) as the primary bottleneck with a Volume-to-Capacity (V/C) ratio of 1.12 (LOS F).
- 5. Optimization Generation:** Marcus clicks "Generate Offline Optimization Schedule." The system processes queuing models and presents a recommended 120-second cycle plan with adjusted green splits favoring the heavily saturated eastbound approach.
- 6. Simulation Evaluation:** System renders side-by-side "before vs. after" metrics: projected average delay drops from 68.4s to 42.1s per vehicle (LOS E to LOS D improvement), and peak queue length drops by 35%.
- 7. AI Rationale Inspection:** Marcus expands the AI Rationale panel, reading: *"Green time for Phase 2 (Eastbound Through) increased by 14 seconds to accommodate morning rush surge observed between 08:00 AM - 08:45 AM. Minimum pedestrian clearance times (Phase 4/8) preserved at 18 seconds."*
- 8. Export & Field Execution:** Satisfied with the results, Marcus exports the NEMA-compliant timing sheet and executive PDF report for field implementation.

15. Functional Requirements

Module A: Data Ingestion & Processing

- **FR-A01:** Ingest structured CSV/JSON file uploads containing hourly or minute-by-minute vehicle counts per intersection approach.
- **FR-A02:** Support manual metadata entry/editing for lane configurations, speed limits, free-flow speeds, and turning movement splits (Left, Through, Right).
- **FR-A03:** Detect and flag corrupted, missing, or outlier data points with options for interpolation or exclusion.
- **FR-A04:** Support optional ingestion of incident logs (accidents, construction events) tagged by timestamp and location.

Module B: Congestion Analytics Engine

- **FR-B01:** Compute Volume-to-Capacity (V/C) ratios for each approach and total intersection per time interval.
- **FR-B02:** Identify peak congestion windows (AM Peak, Midday, PM Peak) automatically using temporal cluster analysis.
- **FR-B03:** Assign Highway Capacity Manual (HCM) Level of Service (LOS) letters (A through F) based on calculated control delay.
- **FR-B04:** Map multi-intersection bottlenecks along defined linear corridors.

Module C: Signal Timing Recommendation Engine

- **FR-C01:** Calculate optimal green split distribution per signal phase based on Webster's method and queue saturation balances.
- **FR-C02:** Calculate recommended total cycle length constrained within user-defined min/max bounds (e.g., 60s to 180s).
- **FR-C03:** Calculate arterial coordination offset values across adjacent intersections to establish green wave progression velocity.
- **FR-C04:** Enforce hard safety lower-bounds for minimum green, yellow change, and red clearance intervals based on road geometry and pedestrian walk speeds.

Module D: Queuing Simulation & Validation

- **FR-D01:** Simulate "before vs. after" queue length accumulation and dissipation using M/M/1 and Webster queuing formulation logic.
- **FR-D02:** Calculate predicted reduction in average control delay per vehicle (sec/veh).
- **FR-D03:** Estimate environmental impact delta (fuel consumption in gallons saved, CO2 emission reduction in kilograms).
- **FR-D04:** Display interactive slider controls allowing users to perform "what-if" traffic volume variation stress testing (+/- 30%).

Module E: AI Rationale & Export Suite

- **FR-E01:** Integrate Azure OpenAI service to generate natural-language explanations for all timing adjustments, detailing cause-and-effect drivers.
- **FR-E02:** Render explicit listing of all simulation parameters, formulas, and assumptions used during calculation.
- **FR-E03:** Generate downloadable NEMA/170/2070 controller-formatted signal timing sheets (CSV format).
- **FR-E04:** Generate branded, client-ready executive summary reports in PDF format.

16. Non-Functional Requirements

- **Performance:** Processing and timing calculation for up to 20 corridor intersections must complete in under 5 seconds. AI narrative generation must return in under 3 seconds.
- **Accuracy & Reliability:** Queuing model delay estimations must align within +/- 10% variance against classical Highway Capacity Manual (HCM) equations under stable flow assumptions.

- **Security:** Role-Based Access Control (RBAC) ensuring data isolation between different municipal accounts. All data in transit encrypted via TLS 1.3; data at rest encrypted via AES-256.
- **Usability:** System must achieve a System Usability Scale (SUS) score above 80. Intuitive design allowing a trained traffic analyst to complete an end-to-end corridor retiming workflow within 5 minutes.
- **Interpretability:** Zero "black box" decisions; every numerical output must link to a readable mathematical formula or explicit AI rationale.
- **Availability:** System web portal uptime availability target of 99.9%.

17. MVP Scope

The Hackathon MVP focuses strictly on providing core offline analytics, signal retiming recommendation, simple simulation, and AI rationale generation for a multi-intersection corridor:

- CSV data ingestion for vehicle volume counts and basic road metadata.
- Peak congestion pattern and bottleneck identification engine.
- Offline green split and cycle length optimization model.
- Simple queuing theory-based "before vs. after" delay and queue length simulation.
- Azure OpenAI powered plain-language rationale generator.
- Interactive Streamlit/Web dashboard displaying metrics, charts, and downloadable PDF/CSV reports.

18. Future Scope

- **Real-Time API Connectors:** Automated data ingestion from city SCADA systems, camera feeds, or cellular probe data (e.g., INRIX/TomTom).
- **Transit Signal Priority (TSP) Integration:** Dedicated optimization modules giving priority green windows to scheduled municipal buses.
- **V2X Communication:** Exporting recommended timing plans via SPaT (Signal Phase and Timing) protocols for connected autonomous vehicles.
- **Multi-Modal Analytics:** Incorporating dedicated bicycle and pedestrian volume counts into dynamic phase allocation algorithms.
- **Deep Reinforcement Learning Simulator:** Integrating SUMO (Simulation of Urban MObility) for heavy microscopic agent-based simulation validation.

19. Feature Prioritization (MoSCoW Matrix)

Category	MUST HAVE (MVP)	SHOULD HAVE	COULD HAVE	WON'T HAVE
Data Ingestion	CSV Upload, Basic Metadata Entry	Batch Ingestion, Anomaly Filter	GIS Shapefile Auto-import	Direct Controller SCADA Stream
Analytics	Peak Window ID, V/C Ratio	Turning Movement Asymmetry	Weather Impact Correlation	Speed Enforcement Profiling
Optimization	Webster Green Splits, Cycle Bounds	Corridor Offset Coordination	Day-of-Week Schedule Banding	Live Closed-Loop Actuation
Simulation	Queuing Theory Delay & Queue	Volume Surge Stress Test Slider	Fuel & CO2 Calculator	Full Micro-Simulation Engine
AI & Reporting	LLM Rationale, PDF/CSV Export	Custom Municipal Branding	Audio Summary Generation	Guaranteed Delay Warranties

20. Success Metrics

- **Time-to-Insight:** Reduction in time required to generate a retiming plan from 14 days (manual baseline) to < 5 minutes.
- **Simulated Delay Reduction:** Average predicted corridor vehicle delay reduction of ≥ 15%.
- **Engineering Trust Score:** ≥ 85% positive rating from traffic engineers regarding the accuracy and clarity of AI rationales.
- **Data Processing Speed:** Parsing and processing 1,000,000 hourly traffic records in < 10 seconds.
- **Export Utilization:** > 90% of completed analysis sessions resulting in a downloaded controller timing sheet or PDF executive summary.

21. Key Performance Indicators (KPIs)

KPI Category	Metric Name	Target Benchmark	Measurement Method
Performance	Corridor Optimization Processing Time	< 5.0 Seconds	System log timestamps
Model Quality	Simulated Queue Reduction Accuracy	+/- 8% vs HCM formulas	Queuing model benchmark suite
User Value	Estimated Vehicle Delay Reduction	12% - 25% per corridor	Before vs. After simulation engine
AI Reliability	Rationale Relevancy & Safety Score	100% MUTCD Safety Compliance	Automated evaluation guardrails
Adoption	Monthly Active Projects per User	≥ 4 Corridors / Month	Analytics platform telemetry

22. Risks & Mitigations

Risk Factor	Impact	Likelihood	Mitigation Strategy
Incorrect Field Implementation	High	Low	Include bold disclaimers, enforce mandatory human engineer sign-off, auto-check clearance intervals.
Inaccurate Input Data	High	Medium	Implement data sanity checks upon upload (flag implausible volumes exceeding saturation flow).
Black-Box AI Hallucination	Medium	Medium	Use structured prompt templates; constrain LLM to explain computed math rather than calculate numbers.
User Resistance to AI	Medium	High	Frame product strictly as an "Engineering Advisory Co-pilot" that augments human expertise.

23. Assumptions

- Historical volume count inputs provided by users reflect underlying demand patterns with reasonable statistical consistency.
- Intersections operate under standard NEMA or fixed-stage phasing sequences that can be mapped into digital timing templates.
- Driver behavior (saturation flow rates, acceleration rates) conforms to baseline Highway Capacity Manual standard constants unless customized by the user.
- Municipal users possess authorization to modify physical controller timing parameters following offline analysis.

24. Constraints & Guardrails

Mandatory Operating Guardrails:

- **Offline Only:** The platform MUST NOT issue direct control commands or interface with live signal infrastructure; all outputs are advisory recommendations.
- **Explicit Assumptions:** All simulation outputs MUST clearly display mathematical models, queuing formulas, and assumed variables used during computation.
- **No Guaranteed Claims:** The system MUST explicitly avoid marketing or reporting claims that guarantee specific real-world travel time reductions, attributing all numbers to simulated estimates.
- **Safety Primacy:** Minimum pedestrian walk times and clearance intervals MUST serve as non-negotiable hard constraints in all optimization routines.

25. Acceptance Criteria

- **AC-01:** System successfully ingests valid multi-intersection CSV traffic count logs and displays clean volume charts without crashing.
- **AC-02:** System correctly identifies the peak 1-hour traffic volume window for any given 24-hour dataset.
- **AC-03:** Calculated green splits sum exactly to the total defined cycle length minus clearance lost times.
- **AC-04:** Recommended green time for any phase never drops below user-defined minimum pedestrian crossing limits.
- **AC-05:** Simulation module calculates and displays both baseline and optimized average delay (sec/veh) and queue lengths (meters) side-by-side.
- **AC-06:** Azure OpenAI integration returns a coherent 2-3 paragraph plain-language narrative explaining recommended green split adjustments.
- **AC-07:** Controller timing export generates a valid, properly formatted CSV file containing phase timing parameters.

26. Competitive Advantage

Traditional incumbents in the traffic engineering space fall into two extremes: expensive, ultra-complex microscopic simulation suites (e.g., VISSIM, Synchro) that require weeks of manual setup, or expensive, real-time adaptive signal hardware systems (e.g., SCATS, SCOOT) requiring millions in hardware overhauls. MetroFlow AI bridges this gap by providing an instant, cloud-native, offline advisory platform that ingests existing volume data, optimizes timing plans via mathematical queuing models, validates results through lightweight simulation, and provides explainable AI rationales—delivering 80% of the value of high-end tools at 5% of the time and cost.

27. Product Differentiators

- **AI-Powered Plain-Language Explanations:** Converts complex mathematical split adjustments into executive-ready narratives that build immediate trust with engineers and city officials.
- **Zero Hardware Dependency:** Works entirely offline with existing historical count data; requires zero capital expenditure on IoT road sensors or new controllers.
- **Instant Queuing Simulation:** Eliminates the multi-week model calibration process required by traditional micro-simulators through automated mathematical queuing validation.
- **Built-In Safety Guardrails:** Hardcodes statutory safety bounds (MUTCD minimum walk times) into the core optimization algorithm so recommendations are legally and operationally sound.

28. Business Value

- **Municipal Cost Reduction:** Reduces signal retiming project costs from \$4,000/intersection to under \$300/intersection, allowing cities to retime corridors annually rather than every 5 years.
- **Economic Productivity Gains:** Reducing corridor travel delay by 15% recovers thousands of lost commuter hours daily, supporting regional economic activity.
- **Environmental Impact:** Decreased idling at mis-timed red lights directly cuts urban fuel waste and greenhouse gas emissions, aiding municipal climate action targets.
- **Streamlined Advisory Workflows:** Enables engineering consulting firms to triple their project throughput without adding headcount.

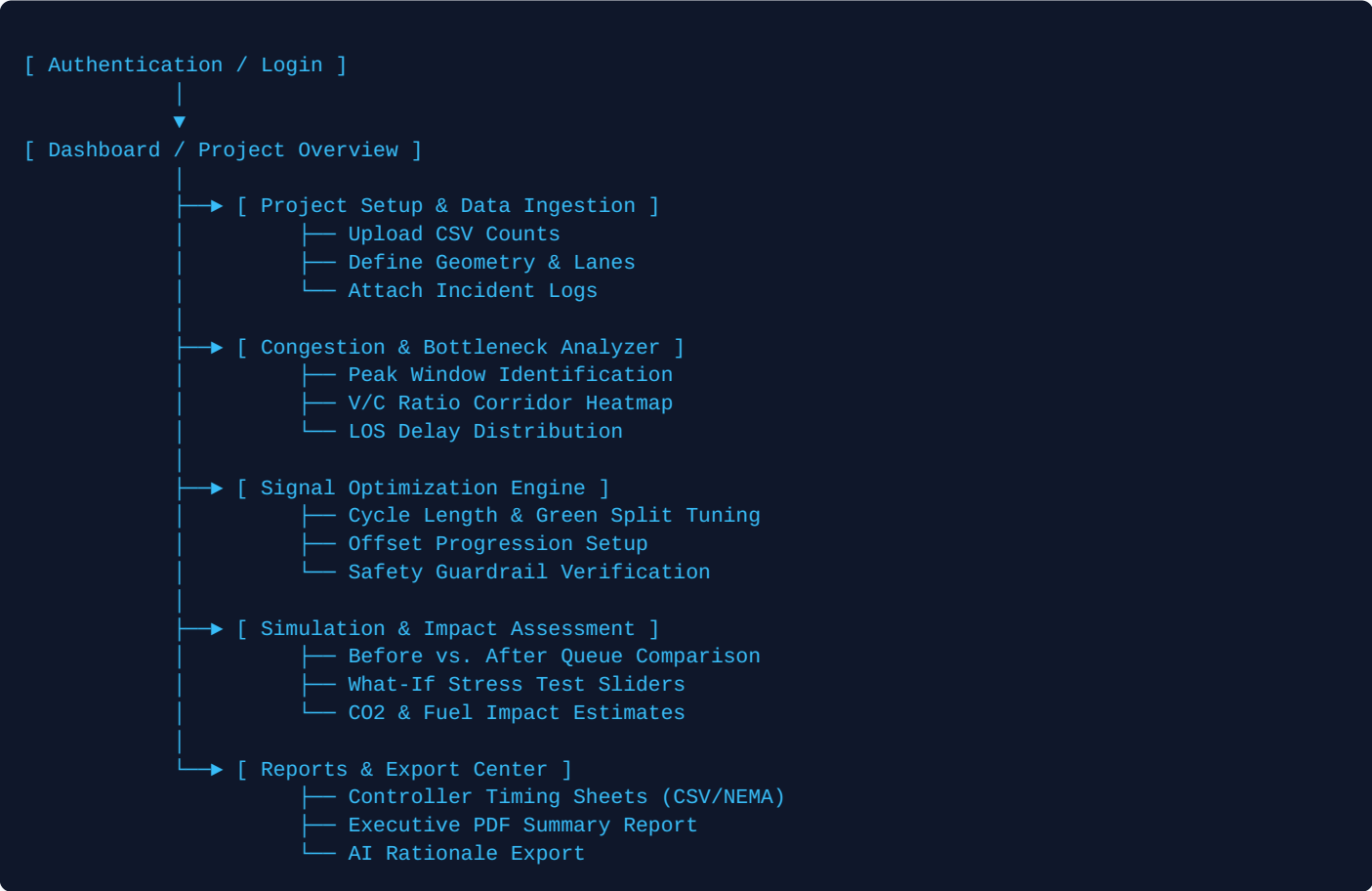
29. Commercialization Potential

- **SaaS Subscription Model:** Tiered annual software subscription for municipal DOTs based on the number of managed intersections (e.g., Small City: \$10,000/yr; Major Metro: \$75,000/yr).
- **Consulting Agency Enterprise Tier:** Tier-1 engineering consultants (e.g., AECOM, WSP, Jacobs) purchase seat-based licenses for client project delivery.
- **Freemium Acquisition:** Free single-intersection analysis tier to drive bottom-up adoption among junior traffic analysts and academic researchers.
- **System Integrator Partnerships:** Partnering with traffic cabinet vendors (e.g., Econolite, Siemens Mobility) to pre-load MetroFlow AI offline analysis suites as value-add software packages.

30. Why Judges Would Choose This Product Over Others

1. **Direct Problem Alignment:** Solves an actual, expensive municipal pain point (static signal retiming) using a realistic, constraint-aware offline methodology.
2. **Responsible AI Application:** Utilizes LLMs where they excel—converting structured numerical analytical results into explainable, clear narrative reports—rather than trusting the LLM to perform mathematical calculations.
3. **Strict Compliance with Constraints:** Meticulously respects every guardrail specified in the problem statement (offline advisory only, explicit simulation assumptions, zero guaranteed claims).
4. **Immediate ROI & Feasibility:** Requires zero real-world infrastructure deployment risk, making it immediately usable by any city DOT tomorrow with their existing CSV data.

Complete Navigation Flow



Screen-by-Screen User Journey

Screen 1: Dashboard & Project Management

- **Header:** Workspace Selector, Search Bar, Municipal Account Profile, System Notifications.
- **Main Area:** Grid view of active retiming projects (e.g., "Downtown Arterial Corridor", "Westside Highway Network").
- **Key Visuals:** Summary cards showing Total Intersections Analyzed, Average Delay Reduced across projects, and Last Plan Update dates.
- **Primary Action:** "+ Create New Corridor Retiming Project" button.

Screen 2: Data Ingestion & Corridor Configuration

- **Left Panel (Upload Area):** Drag-and-drop box for traffic count CSV files and incident logs. Automated file validation indicator.
- **Right Panel (Geometry Builder):** Interactive node map where users select intersections, specify approach lane counts, speed limits, and turning ratios.
- **Bottom Bar:** "Parse & Analyze Congestion Data" primary action button.

Screen 3: Congestion Analytics & Bottleneck Dashboard

- **Top Ribbon:** Metric cards showing Peak Hour Identified (e.g., 08:00 AM - 09:00 AM), Worst Intersection V/C Ratio (1.14), and Corridor Average Delay (54.2s/veh).
- **Center Map / Heatmap:** Color-coded arterial corridor map highlighting red/orange bottleneck intersections.
- **Bottom Charts:** Hourly volume bar charts and approach-level Level of Service (LOS) breakdown graphs.
- **Primary Action:** "Proceed to Signal Optimization" button.

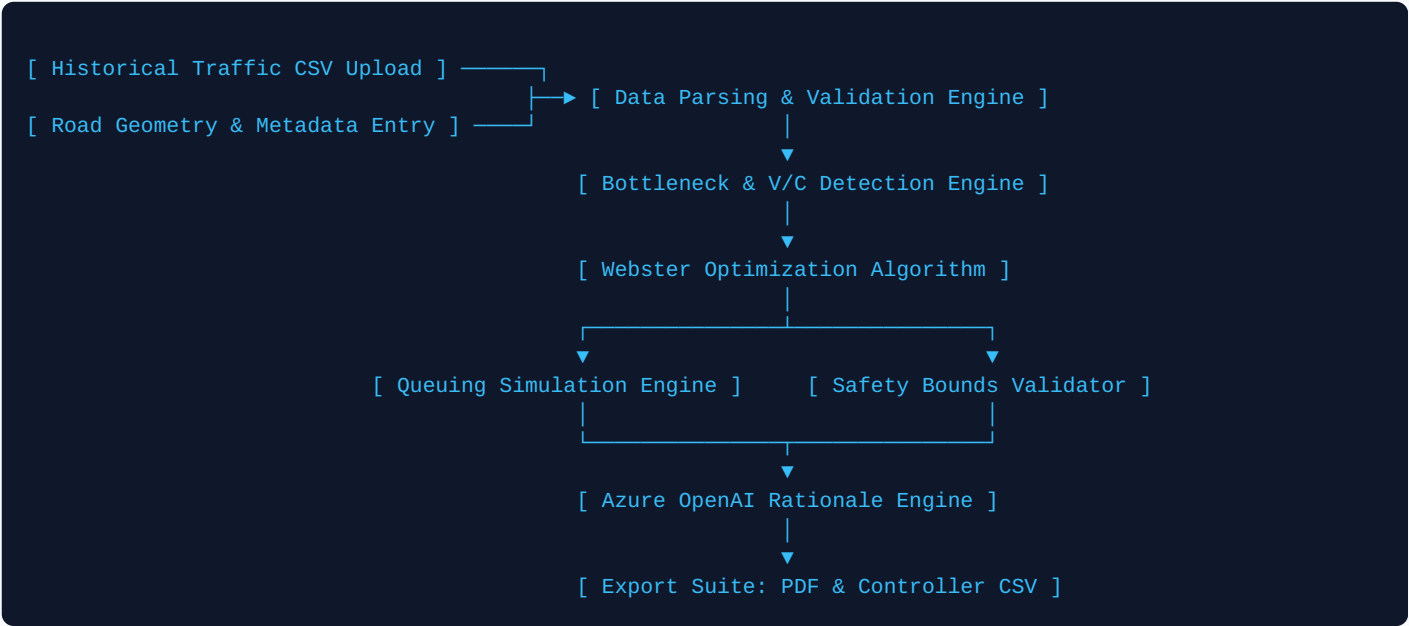
Screen 4: Optimization & Simulation Workbench

- **Split View Layout:**
 - **Left Side (Timing Parameters):** Interactive phase diagram and slider controls for Cycle Length (90s - 150s), Phase Green Splits (seconds), and Offset values. Displays real-time green time validation warnings if safety walk times are breached.
 - **Right Side (Simulated Results):** Side-by-side "Before vs. After" comparative metrics: Control Delay (54.2s → 36.1s), Queue Length (120m → 75m), Intersection LOS (E → C).
- **Bottom Controls:** Interactive "Volume Surge Stress Test" slider (+/- 30% traffic variation).

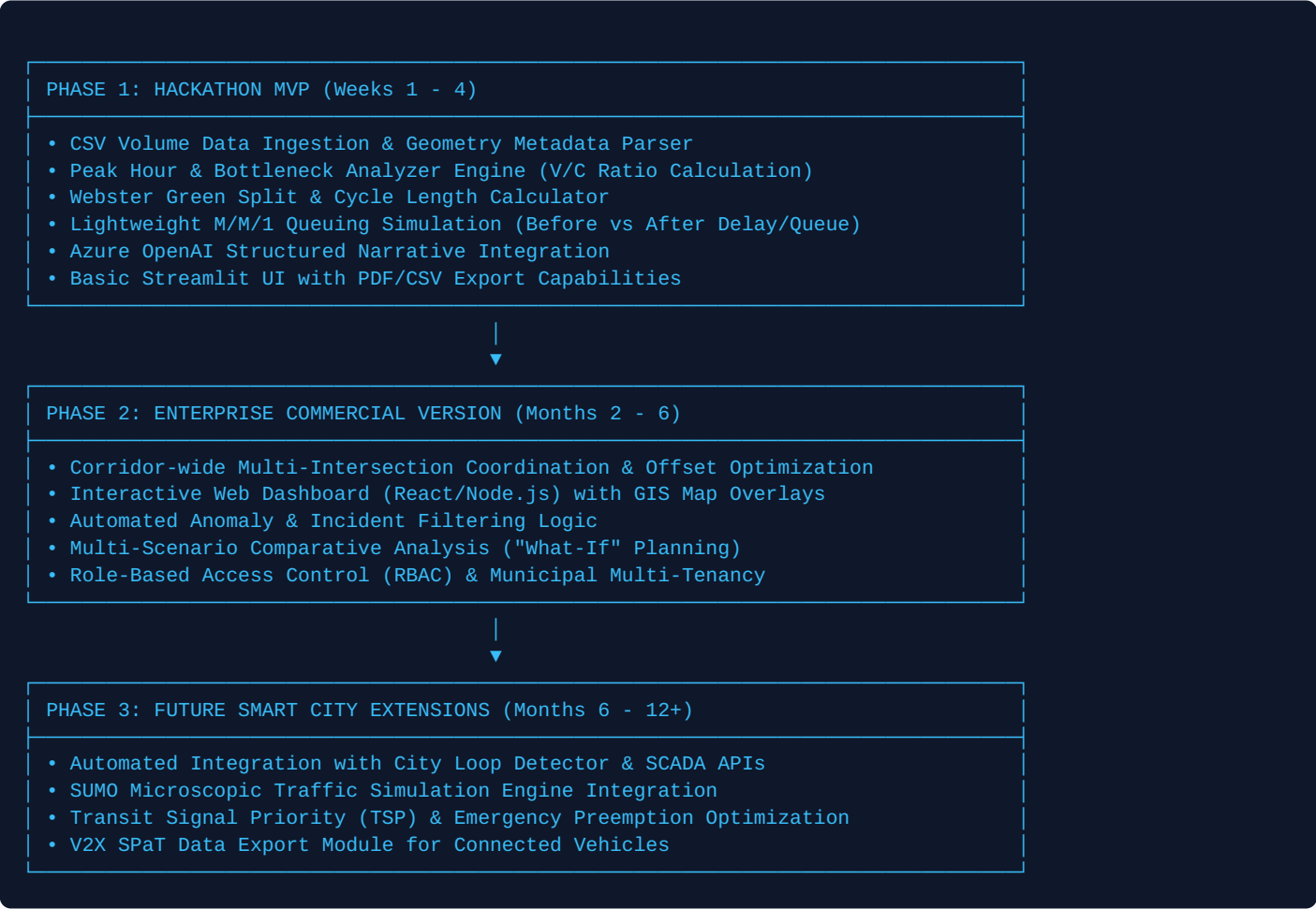
Screen 5: AI Rationale & Report Export Center

- **Left Panel (AI Narrative):** Markdown-formatted narrative generated by Azure OpenAI explaining the engineering justification for split shifts, cited standards, and explicit simulation assumptions.
- **Right Panel (Export Preview):** Live preview of the Controller Parameter Table (NEMA format) and Executive PDF Summary.
- **Primary Actions:** "Download Controller Timing Sheet (CSV)", "Download Executive PDF Report", "Save & Lock Project Schedule".

Feature Dependency Map



Product Roadmap



Suggested Product Names (20 Options)

- | | |
|-----------------------------------|------------------------|
| 1. MetroFlow AI (Selected) | 1. MetroSignal AI |
| 2. SignalGrid AI | 2. UrbanFlow Studio |
| 3. UrbanPulse Mobility | 3. SignalPulse Engine |
| 4. RetimeIQ | 4. CorridorIQ |
| 5. GridOptima | 5. FlowPhase Analytics |
| 6. FlowSync AI | 6. OptiGrid Urban |
| 7. ClearPhase AI | 7. GridSync AI |
| 8. TrafficLogic Pro | 8. TrafficMind Offline |
| 9. SmartSplit AI | 9. UrbanRetime AI |
| 10. TransitMatrix | 10. ApexTraffic AI |

Suggested Taglines (20 Options)

1. *Unlocking urban movement through explainable signal retiming.* (Selected)
2. *Smarter signals, smoother corridors, zero live risk.*
3. *Turning historical traffic data into intelligent signal decisions.*
4. *Eliminate city gridlock, one phase at a time.*

5. *Offline signal optimization powered by transparent AI.*
6. *Data-driven traffic engineering made fast and explainable.*
7. *Transforming traffic counts into optimized green light schedules.*
8. *Engineering better urban flow without new hardware.*
9. *From raw volume logs to optimized signal timing in minutes.*
10. *Precision corridor tuning for modern smart cities.*
11. *Simulate, optimize, and retune with complete confidence.*
12. *Say goodbye to static signal timing gridlock.*
13. *The intelligent advisory co-pilot for municipal traffic engineers.*
14. *Optimize green splits, reduce delay, lower city emissions.*
15. *Clear visual analytics and safe offline signal recommendations.*
16. *Bridging traditional traffic engineering with explainable AI.*
17. *Intelligent signal retiming without the micro-simulation overhead.*
18. *Smarter green split allocation backed by queuing science.*
19. *Unlocking arterial throughput using the data you already have.*
20. *Safe, simulated, and optimized traffic management for modern cities.*

Critical Evaluation & Judge Review

1. Key Product Weaknesses

- **Reliance on Static Historical CSVs:** The MVP assumes municipal count data is clean and structured. In reality, city loop detector logs are notoriously messy, corrupted, or contain long gaps.
- **Simplified Queuing Model Limitations:** M/M/1 and Webster queuing models assume steady-state arrivals, which can under-predict delay during extreme oversaturated conditions ($V/C > 1.2$) where queues compound exponentially.
- **Lack of Manual Override Granularity in MVP:** Early UI designs focus heavily on automated AI generation, which can alienate conservative traffic engineers who demand granular control over individual phase clearance intervals.

2. Missing Features

- **Pedestrian & Bicycle Phasing Constraints:** The initial feature set emphasizes vehicular throughput, but urban traffic engineers must legally accommodate minimum pedestrian walk and flashing don't walk (FDW) intervals.
- **Corridor Offset Progression Diagrams (Time-Space Diagrams):** Traffic engineers rely heavily on Time-Space diagrams to visualize green wave bands across multi-intersection corridors. This visual was omitted from early functional requirements.

3. Unnecessary / Over-Engineered Features

- **Complex LLM Natural Language Chat Interfaces:** Adding a conversational chatbot for retiming setup adds unnecessary complexity and non-deterministic behavior. A structured narrative summary report is far more practical for municipal compliance.
- **Real-time API Push Intentions:** Attempting to build real-time signal controller integration hooks violates problem statement guardrails and creates severe municipal security/safety liability concerns.

4. Opportunities to Improve Before UI/UX Design

- **Integrate HCM Minimum Safety Hard-Bounds:** Hardcode statutory minimum pedestrian crossing formulas explicitly into the optimization algorithm so the UI automatically blocks invalid green split adjustments.
- **Add Interactive Time-Space Progression Diagrams:** Include a visual Time-Space graph on Screen 4 showing green wave bands along the corridor.

- **Implement Automated CSV Data Cleaning:** Add an interactive data-cleaning step during ingestion that highlights and repairs missing timestamps or abnormal volume spikes before running calculations.
- **Focus Heavy UI Real-Estate on "Before vs. After" Validation:** Design Screen 4 around a side-by-side comparative layout that clearly highlights delay reduction and queue improvements to give engineers immediate confidence in the recommendation.